SIMATS ENGINEERING

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Jotheesh A
192511029RD

Class Work(28.07.26)

← Back

QUESTIONS

Q1
Void Function (No Return Value)
5 marks

Q2
Void Function (No Return Value)
5 marks

Q3
Void Function (No Return Value)
5 marks

Q4
Fruitful Function (Returns One Value)
5 marks

Q5
Fruitful Function (Returns One Value)
5 marks

Q6
Fruitful Function (Returns One Value)

Q1 **Void Function (No Return Value)**

5 marks

Write a Python program to define a function that displays "Welcome to Python Programming".

Your Code

```
def welcome():
    print("Welcome to Python Programming")
welcome()
```

Input

```
Welcome to Python Programming
```

Output

```
Welcome to Python Programming
```

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Q6 Fruitful Function (Returns One Value)

Q2 **Void Function (No Return Value)** 5 marks

Write a function to print the multiplication table of a given number.

Your Code

```
def multiplication_table(n):
    for i in range(1, 11):
        print(f"{n} x {i} = {n * i}")

n = int(input())
multiplication_table(n)
```

Input

```
6
```

Output

```
6 x 1 = 6
6 x 2 = 12
6 x 3 = 18
6 x 4 = 24
6 x 5 = 30
6 x 6 = 36
```

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Q6
Fruitful Function (Returns One

Q3 **Void Function (No Return Value)**

5 marks

Write a function to print all even numbers from 1 to N.

Your Code

```
def print_even(n):
    for i in range(2, n + 1, 2):
        print(i)

n = int(input())
print_even(n)
```

Input

```
4
```

Output

```
2
4
```

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5 marks

Q6
Fruitful Function (Returns One Value)

Q4 **Fruitful Function (Returns One Value)** 5 marks

Write a function that accepts a string and returns the number of vowels in it.

Your Code

```
def count_vowels(s):
    vowels = "aeiouAEIOU"
    count = 0
    for char in s:
        if char in vowels:
            count += 1
    return count

s = input()
print(count_vowels(s))
```

Input

```
Hello World
```

Output

```
3
```

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- Q5 Fruitful Function (Returns One Value) 5 marks
- Q6 Fruitful Function (Returns One

Q5 **Fruitful Function (Returns One Value)** 5 marks

Write a function that accepts a list of integers and returns the largest element.

Your Code

```
def find_largest(lst):
    return max(lst)

numbers = list(map(int, input().split()))
print(find_largest(numbers))
```

Input

```
2 4 5 8 9 6 12
```

Output

```
12
```

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Q5
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5 marks

Q6
Fruitful Function (Returns One Value)

Q6 **Fruitful Function (Returns One Value)** 5 marks

Write a function that accepts a number and returns whether it is prime or not.

Your Code

```
def is_prime(n):
    if n < 2:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True

n = int(input())
print(is_prime(n))
```

Input

7

Output

True

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Q2
Void Function (No Return Value)
5 marks

Q3
Void Function (No Return Value)
5 marks

Q4
Fruitful Function (Returns One Value)
5 marks

Q5
Fruitful Function (Returns One Value)
5 marks

Q6
Fruitful Function (Returns One Value)
5 marks

Q7
Function Returning Two or More Values
5 marks

Q8
Function Returning Two or More Values

Your Code

```
def fib_recursive(n, a=0, b=1, terms=0):
    if terms == n:
        return []
    return [a] + fib_recursive(n, b, a + b, terms + 1)

n = int(input())
print(*fib_recursive(n))
```

Input

```
9
```

Output

```
0 1 1 2 3 5 8 13 21
```

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Q6
Fruitful Function (Returns One Value)
5 marks

Q7

Q8 Function Returning Two or More Values

5 marks

Write a function that accepts the radius of a circle and returns its area and circumference.

Your Code

```
import math

def circle_area_and_circumference(r):
    area = math.pi * r * r
    circumference = 2 * math.pi * r
    return area, circumference

# Example
r = float(input("Enter radius: "))
a, c = circle_area_and_circumference(r)
print("Area:", a)
print("Circumference:", c)
```

Input

```
5
```

Output

```
Enter radius: Area:
78.53981633974483
Circumference: 31.41592653589793
```

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5 marks

Q6
Fruitful Function (Returns One Value)

Q9 **Function Returning Two or More Values**

5 marks

Write a function that accepts a number and returns its square, cube, and square root.

Your Code

```
import math

def square_cube_sqrt(n):
    return n**2, n**3, math.sqrt(n)

n = float(input("Enter a number: "))

sq, cu, sr = square_cube_sqrt(n)

print(sq, cu, sr)
```

Input

```
6
```

Output

```
Enter a number: 36.0 216.0
2.449489742783178
```

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5 marks

Q6
Fruitful Function (Returns One Value)

Q10 **Keyword Arguments**

5 marks

Write a function student(name, age, department) and call it using keyword arguments.

Your Code

```
def student(name, age, department):
    print("Name:", name)
    print("Age:", age)
    print("Department:", department)

student(name="Jotheesh A", age=20, department="CSE")
```

Input

stdin input

Output

```
Name: Jotheesh A
Age: 20
Department: CSE
```

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    print("Department:", department)

student(name="Jotheesh A", age=20, department="CSE")
```

Input

stdin input

Output

```
Name: Jotheesh A
Age: 20
Department: CSE
```

Submitted